# Department of Computer and Software Engineering

# Lab Project

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## Submitted To:

**SIR ASIM MUNIR**

**Title: Access Control System Room Security and Automation System Design**

**Introduction:** This project focuses on designing a system that integrates comparison logic, counter functionality, and data handling using digital logic design principles. The system is developed using Multisim simulation software, ensuring accurate and reliable implementation. The circuit utilizes DIP switches for input, comparators for binary value comparison, shift registers for data manipulation, and counters for digital counting and display.

**Requirements / Specifications:**

- **Input Voltage:** 5 V DC logic levels
- **Logic Components:** AND, OR, NOR gates, and shift registers
- **Outputs:** Digital display for counter values and comparator results
- **Functional Validation:** Circuit performance is verified using simulations in Multisim

**Components and Their Functions:**

**DIP Switches (4 units):**

- Allow manual binary input to the system.
- Each switch corresponds to a single bit of the binary input.

**7485N (4 units - Digital Comparators):**

- Function as 4-bit binary comparators.
- Compare two 4-bit binary inputs and provide equality, greater-than, and less-than outputs.

**CNTR_4ADEC (2 units - 4-bit Counters):**

- Implement counters with outputs connected to seven-segment displays.
- Display the results of the counting process.

**74194N (4 units - 4-bit Shift Registers):**

- Perform shifting, parallel load, and storage of binary data.
- Allow flexibility in manipulating input data for further operations.

**Logic Gates:**

- **AND Gates:** Combine comparator outputs to validate specific conditions.
- **OR Gates:** Merge multiple logical conditions for system functionality.
- **NOR Gates:** Provide inverted logic outputs for necessary circuit operations.

**Power Sources (5V):**

- Provide the necessary DC logic levels for the circuit operation.

**Circuit Design and Working:**

**Comparison Logic:**

- Each 7485N IC compares two 4-bit inputs.
- Outputs are used to determine equality, greater-than, or less-than conditions for the inputs.

**Counter and Display:**

- CNTR_4ADEC counters increment or decrement based on input conditions.
- The counter outputs are connected to seven-segment displays for visual representation.

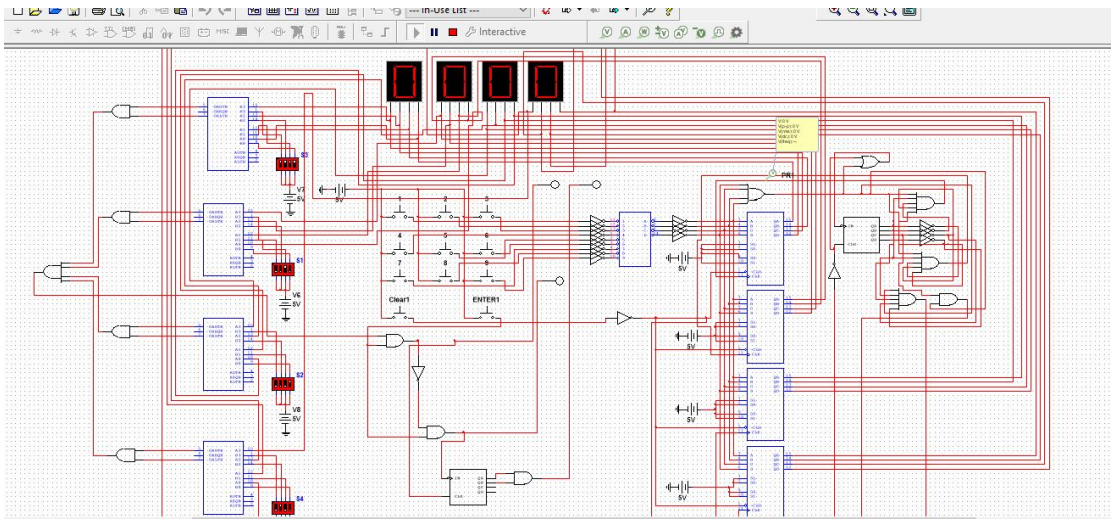
**Data Manipulation:**

- 74194N shift registers handle parallel data loading and shifting operations.
- Data manipulation is synchronized with system logic for accuracy.

**Control Logic:**

- Logic gates are used to control the system's operation and validate specific conditions.
- The design ensures reliable operation and accurate output generation.

**Simulation Results:**



**Observations and Outputs:**

- The comparators correctly identified equality, greater-than, and less-than conditions.
- The counters displayed the correct binary count on the seven-segment displays.
- Shift registers accurately manipulated data according to input logic.

**Conclusion:** The project successfully demonstrates the integration of comparators, counters, and shift registers in a digital logic system. The circuit was designed and simulated using Multisim, with all functionalities verified through rigorous testing. This project highlights the practical application of digital logic design principles in creating reliable and efficient systems.